CSI 4170 / 5170: Machine Learning — Final Project Phase 1

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1. Project Identification

Project Title: Data Quality and Model Performance: Comparing Classifier Behavior on a Clinical Benchmark and a Self-Collected Daily Log

2. Groups

Student 1: Jing Yee — jyee@oakland.edu
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3. Project Specifications

Primary Goals, Aims, and Intended Outcomes:
To evaluate whether standard classification algorithms behave differently when applied to a large, clean benchmark dataset versus a small, self-collected dataset. Both datasets are processed through an identical pipeline so that differences in performance can be attributed to data characteristics (e.g. sample size, noise, and reporting subjectivity) rather than to differences in method. A second objective is to determine how much of any performance gap can be recovered through data refinement: feature scaling, outlier removal, feature selection and hyperparameter tuning are applied as a cumulative sequence, and the metrics are recorded at each stage so the value of each refinement is measured rather than assumed.

Specific Problem Addressed:
Personal wellness applications frequently fail in practice because machine learning models trained on large public survey datasets underperform when applied to subjective, self-logged daily human habits.

Key Features, Functionalities, and Deliverables:

- A unified data cleaning and feature engineering pipeline built with pandas and scikit-learn.
- Exploratory data analysis of each dataset separately: summary statistics, feature distributions, and correlation heatmaps, compared in terms of structure rather than shared variables.
- Three classifiers, Naive Bayes, Logistic Regression and a Support Vector Machine, trained on each dataset against its own binary target: malignant versus benign for the benchmark data, and high versus low stress for the self-collected data. Each is evaluated before and after a staged sequence of data refinements, so the contribution of each refinement step can be measured.

- Evaluation metrics comparing performance on both datasets across Accuracy, Precision, Recall, F1 Score, and Confusion Matrices.

Scope and Boundaries:

- **Included:** Data collection schema design, data preprocessing, feature scaling, exploratory analysis, model training, hyperparameter tuning, and metric evaluation on both datasets.
- **Excluded:** Developing live mobile applications, creating custom wearable hardware, or setting up real-time server database synchronization.

4. Motivation and Problem Statement

Problem Description:
Daily lifestyle choices (sleep, caffeine intake, screen time, exercise) are associated with physiological stress and overall well-being. However, personal lifestyle data logged by individuals contains significant human noise, missing observations, and subjective reporting bias compared to data collected under controlled survey conditions.

Importance:
Most published model performance is reported on large, carefully curated datasets. Real applications often run on small, self-reported data collected under far less controlled conditions. Quantifying how much performance degrades under those conditions is useful for anyone deploying a model outside a research setting.

5. Tasks Management

5.1 Timeline (Gantt Chart Table)

Task / Phase	Oct 6–12	Oct 13–19	Oct 20–26	Oct 27–Nov 2	Nov 3–9	Nov 10–16	Nov 17–23	Nov 24–30	Dec 1–7
Phase 1 proposal (due Oct 16)	■	■							
Daily habit logging — all 3 members, ongoing	■	■	■	■	■	■	■	■	
Literature review & related work			■	■					
Data preprocessing, scaling & EDA			■	■					
Model implementation & hyperparameter tuning				■	■	■			
Pipeline diagram & Phase 2 design document (due Nov 13)					■	■			
Model evaluation & performance metrics						■	■		

Task / Phase	Oct 6–12	Oct 13–19	Oct 20–26	Oct 27–Nov 2	Nov 3–9	Nov 10–16	Nov 17–23	Nov 24–30	Dec 1–7
GitHub repository & data publication setup									
Final report & presentation (Phase 3, due Dec 4)									

Shaded cells indicate the weeks in which each task is active. Task dependencies: the logging schema must be finalized before daily collection can begin; collection must be underway before any evaluation on the self-collected data is possible; and the evaluation results are required before the Phase 3 report can be written. Milestones: Phase 1 due October 16, Phase 2 due November 13, Phase 3 due December 4.

5.2 Responsibilities

- **Jing Yee (Data Engineer):** Responsible for designing the shared Google Sheet logging schema, managing the preprocessing pipeline, handling missing values, and applying feature scaling.
- **Pablo Avila (Machine Learning Developer):** Responsible for building, training, and tuning the classification models (Naive Bayes, Logistic Regression, Support Vector Machine) across both datasets, and running the staged refinement experiments.
- **Christina Carvalho (Evaluation & Report Lead):** Responsible for exploratory data visualisations (correlation heatmaps), calculating evaluation metrics (Accuracy, Precision, Recall, F1, Confusion Matrices), and authoring final project documentation.

6. Data Availability

Process for Obtaining Data:
Public benchmark data will be imported directly using scikit-learn.datasets (load_breast_cancer, the Wisconsin Diagnostic Breast Cancer dataset). This dataset was selected because it is distributed in raw, unprocessed form: its features span very different numeric ranges, which makes it suitable for demonstrating the effect of preprocessing on model performance. It is natively a two-class problem, so no artificial thresholding of the target is required. Personal behavioral data will be logged daily into a shared Google Sheet by all 3 team members continuously from the agreed start date through the end of Phase II, yielding approximately 150 observations.

Types of Data Required:
Continuous numerical input features (sleep_hours, exercise_mins, screen_time_hours, study_hours, caffeine_cups) paired with 1–10 ordinal scale target indicators (mood, stress, focus). For classification tasks, targets will be threshold-binned into binary categories (e.g., High Stress 6 or above vs. Low Stress 5 or below). The benchmark dataset provides 569 records with 30 numeric features computed from digitized images of cell nuclei (radius, texture, perimeter, area, smoothness and related measurements), paired with a binary diagnosis label (malignant

or benign).

Accessibility & Quality Challenges:

- Limited sample size (approximately 150 rows) for the self-tracked group dataset compared to larger public datasets.
- Subjective self-reporting variance and personal rating calibration across team members rating 1–10 stress/mood scales.
- Behavioral variations between weekdays and weekends.

Impact on Model Performance:

Higher noise, subjective variance, and smaller sample size in self-logged data may lead to lower classification accuracy or increased susceptibility to overfitting compared to the standardized, clean clinical benchmark dataset.

7. AI Usage Statement

Claude has been used to help organize and draft the report. The final output retains the core thought processes of the students and has been reviewed by the group.

8. References

1. Pedregosa, F., et al. (2011). Scikit-learn: Machine Learning in Python. Journal of Machine Learning Research, 12, 2825–2830.
2. Street, W. N., Wolberg, W. H., & Mangasarian, O. L. (1993). Nuclear feature extraction for breast tumor diagnosis. IS&T/SPIE International Symposium on Electronic Imaging: Science and Technology, 1905, 861–870. (Source of the scikit-learn breast cancer dataset.)
3. Huang, Y., Yang, L., Liu, Y., & Zhang, S. (2024). Effects of perceived stress on college students' sleep quality: A moderated chain mediation model. BMC Psychology, 12, 476. https://doi.org/10.1186/s40359-024-01976-3
4. Bradley, A. H. M., & Howard, A. L. (2023). Stress and mood associations with smartphone use in university students: A 12-week longitudinal study. Clinical Psychological Science, 11(5). https://doi.org/10.1177/21677026221116889
5. Anthropic. (2026). *Claude* [Large language model]. https://claude.ai